

NGUYEN QUOC DAI

Software Engineer

Ho Chi Minh City, Vietnam | dainguyen020199@gmail.com | +84 961 664 550

PROFESSIONAL SUMMARY

Software Engineer with over 4 years of hands-on experience designing and delivering scalable, production-grade web applications using modern TypeScript stack. Strong computer science fundamentals with consistent application of OOP, SOLID principles, and clean architecture. Specialized in Next.js 14 App Router, NestJS microservices, interactive data visualization (Highcharts Stock, ECharts), and end-to-end system ownership from architecture to deployment.

CORE TECHNICAL SKILLS

Languages: TypeScript, JavaScript, SQL, C#

Frontend: Next.js, React.js, Redux Toolkit, Vue.js, Nuxt, React Native

Backend: NestJS, Node.js, Express, ASP.NET Core, Redis

Databases: PostgreSQL (advanced indexing & partitioning), MySQL, MongoDB

Data Visualization: Highcharts (Stock, Gantt, custom modules), ECharts, ApexCharts

DevOps & Cloud: Docker, Nginx, GitHub Actions CI/CD, AWS (S3, EC2, ECS, Lambda), Linux

Others: WebSocket, Firebase, Jest, Postman, Jira, Storybook

PROFESSIONAL EXPERIENCE

FIDT Joint Stock Company (Financial)

June 2025 – Present

Role: Full-Stack Software Engineer

Project: WFM – A FinTech platform, delivering personal financial management and real estate investment advisory services to thousands of users.

- Contributed to the development of a comprehensive wealth management platform, providing rapid financial health assessments and data-driven real estate investment recommendations.
- Improved frontend performance through image preloading, asset caching, and hybrid SSR/SPA rendering, resulting in noticeably faster page load times and better user experience.
- Integrated BizFly CRM to support: Sales team commission processing, Deal pipeline tracking
- Technologies: TypeScript, Node.js, Directus, Express, Vue3, Nuxt4, Playwright, Tailwind, Hono

Project: IAM – Is a identity and Access management of FIDT

- Participate in project development and be in charge of tasks related to user authorization processing, authentication - token verification at API Gateway, ensuring data security
- Deploy middleware mechanism to verify tokens according to IAM system standards
- Build and optimize authorization logic by role/group for each endpoint in the system.
- Technologies: TypeScript, Node.js, Directus, Express, Vue3

Project: ER – Is a real estate project, providing customers with an overview of each property.

- Work with the Product Owner to finalize the project structure and features
- Participate in project development according to the FIDT framework
- Responsible for issues related to SEO, testing, performance
- Technologies: TypeScript, Node.js, Directus, Express, Vue3

Vietstats Co., Ltd (Economic Data Analytics & AI Platform)

Sep 2023 – June 2025

Role: Full-Stack Software Engineer (promoted to Leader in 2025)

Project: Data BOT – AI-Powered Economic Data Agent – Intelligent assistant powered by GPT-4 that answers complex economic queries in natural language

- Designed and shipped multi-agent system combining GPT-4 with custom tools (search, comparison, time-series retrieval, statistical functions, ...)
- Implemented iterative reasoning loop with supervisor agent for data validation and refinement until answer quality threshold is met
- Integrated real-time context injection from internal economic databases and live Kafka streams
- Built tool-calling framework and prompt engineering pipeline → achieved 92%+ accuracy on internal benchmark Technologies: TypeScript, Node.js, OpenAI GPT, LangChain-style, PostgreSQL, Docker

Project: portal.vietstats.vn – Provides Data Visualization tools → Allows media agencies and businesses to configure and embed interactive economic charts into other CMSs

- Built configuration dashboard with Next.js 14 App Router + React Hook Form, reducing chart creation time from hours to under 3 minutes
- Developed secure embed script (iFrame + postMessage) and REST/GraphQL APIs with JWT, rate-limiting, and query caching
- Implemented real-time data streaming via WebSocket + Kafka and advanced Highcharts Stock visualizations (drill-down, dynamic themes, annotations)
- Containerized entire platform with Docker Compose, configured Nginx reverse proxy, and automated zero-downtime deployments via GitHub Actions
- Technologies: TypeScript, Next.js 14, NestJS, PostgreSQL, Highcharts Stock, ECharts, Kafka, Docker, Nginx

Project: Advance – Advanced Analytics Workspace for C-level & Analysts

Single-handedly architected and delivered a “Photoshop-like” data workspace: drag-and-drop multi-chart layouts, real-time collaboration, custom indicators, saved templates, export to PDF/PNG and more

- Designed component-based design system, project scaffolding, backend microservices, and entire CI/CD pipeline

- Implemented complex canvas engine with resizable grid, synchronized state across tabs, and live data streaming via Kafka + WebSocket
- Technologies: TypeScript, Next.js 14, NestJS, PostgreSQL, Highcharts, ECharts, Kafka, Docker, Nginx

Project: vietstats.vn – Data-as-a-Service platform providing Vietnam’s economic-level data.

- Architected from scratch the entire platform: PostgreSQL schema with partitioning for multi-million-row datasets, NestJS microservices, and Next.js 14 SSR frontend
- Designed system architecture and database schema aligned with business logic
- Built secure RESTful + GraphQL APIs with role-based access, query caching, and rate limiting
- Containerized services with Docker, configured Nginx + multi-region setup, and automated CI/CD via GitHub Actions

Doctor Check JSC (*Digital Healthcare Provider*)

Mar 2021 – Aug 2023

Role: Full-Stack & Frontend Engineer (promoted to Full-Stack in 2022)

Project: Doctor Check Customer Portal – Patient-Facing Web Platform

- Led development of end-to-end patient portal (Next.js + NestJS + SQL Server) for appointment booking, medical reports, and payment
- Reduced average page load time by 60% through SSR, code splitting, and image optimization
- Technologies: Next.js, NestJS, SQL Server, Ant Design, Redux Toolkit

Project: Doctor Check CRM & Omni-Channel Messaging System

- Built real-time CRM dashboard integrating Facebook Messenger & Zalo OA via WebSocket for 50+ agents
- Implemented customer timeline, tagging, and automated workflows → improved agent response time by 45%
- Technologies: React.js, TypeScript, Redux Toolkit, Firebase, WebSocket

Project: Zalo Mini App – Healthcare Mini App inside Zalo Super App

- Single-handedly designed, developed, and shipped appointment & medical result app reaching 100K+ patients
- Authored UI/UX guidelines and technical documentation adopted company-wide
- Technologies: React.js, TypeScript, Firebase

Project: Doctor Check HIS – Hospital Information System (Internal Tool)

- Migrated legacy modules to modern React SPA architecture, improving internal staff productivity by 30%
- Technologies: Typescript, React.js, MUI, WebSocket

EDUCATION

- Bachelor of Information Technology

- Ho Chi Minh City Industry and Trade College (2018 – 2021)

ACHIEVEMENTS

- Top 15 National Finalist – Startup Kite 2021 (Vietnam’s largest startup contest)
- National Encouragement Award – AI-based smart attendance system with automatic temperature detection

LANGUAGES

Vietnamese (Native) | English (Full professional proficiency)